Curated Research Articles

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- **Ligand Electronics and Metal-Site Acidity Enable High-Voltage MOF Electrolytes for Quasi-Solid-State Lithium-Metal Batteries** — score: 1.000 The study introduces Al-FBDC, a metal-organic framework (MOF) electrolyte that combines electron-withdrawing ligands and acidic metal sites to enhance high-voltage stability in quasi-solid-state lithium-metal batteries (LMBs). By integrating Al-FBDC into NCM622 composite cathodes, the electrolyte effectively reduces decomposition and interfacial degradation, leading to improved cycling stability and performance at high voltages. This research establishes a design framework for developing durable high-energy-density LMBs suitable for demanding applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electrolyte-Derived Chemical Activation Enables Reversible Cathode Chemistry in Magnesium Batteries** — score: 1.000 This study introduces an innovative approach to enhance the electrochemical performance of rechargeable magnesium (Mg) batteries by utilizing trace water in the electrolyte to activate cathodes. By promoting the formation of magnesium/chlorine species, the method improves cathode reversibility in various materials, including MnO and organic compounds, while also preventing unwanted reactions at the magnesium anode. This electrolyte-focused activation strategy offers a new avenue for optimizing magnesium battery chemistry. Journal: *ACS Energy Letters ASAP*
- **Nanostructure Engineering of Copper Sulfide Nanosheets Enables Reversible and Kinetically Boosted Magnesium Storage** — score: 1.000 The article discusses the development of a two-dimensional copper sulfide (CuS) nanosheet cathode for rechargeable magnesium batteries, which addresses challenges such as slow magnesium ion diffusion and structural degradation in traditional materials. It details how the nanosheet design improves reversible capacity and electrochemical performance through enhanced ion migration and charge transfer, while elucidating a multi-step conversion reaction mechanism that aids in structural stability during charging and discharging cycles. Journal: *ACS Applied Energy Materials ASAP*
- **Ion–Dipole Interaction Engineering in Weakly Solvated Polymer Electrolytes for Stable Li-O 2 Batteries** — score: 1.000 This article presents a novel gel polymer electrolyte (GPE) for lithium-oxygen (Li-O2) batteries, designed by integrating a weakly solvating siloxane network with a polar carbonate matrix through in situ copolymerization. The resulting electrolyte, poly(VEC-co-MAPTMS), shows improved ionic conductivity and electrochemical stability, enabling enhanced lithium deposition and a significant extended cycle life for the battery, addressing key challenges in high-energy-density applications. Journal: *ACS Applied Energy Materials ASAP*
- **Electric Double Layer-Driven Surface Reconstruction Enables Cooperative Zn(002)/(101) Growth for High-Utilization Zinc Anodes** — score: 1.000 The study introduces imidazolidinyl urea (IDU) as a solution to improve the performance of aqueous zinc-ion batteries by facilitating selective crystal facet regulation on zinc surfaces. This manipulation enhances the plating and stripping process while minimizing parasitic reactions, allowing cells to operate stably for extended cycles and under various conditions, ultimately improving the battery's efficiency and longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Bridging Ionic Field Strength Disparity in O3-Type Layered Oxides for High-Voltage Sodium-Ion Batteries** — score: 1.000 This article introduces a new ionic field strength disparity coefficient ( ) to evaluate the local chemical-field heterogeneity in O3-type layered oxides used as cathodes in sodium-ion batteries. By incorporating compensatory cations, the disparity is minimized, leading to improved structural stability and electrochemical performance, as demonstrated by a substantial capacity retention of over 80% after 500 cycles at 10C, compared to just 22% for unmodified samples. This study highlights a novel approach for enhancing the durability of high-voltage battery materials. Journal: *Journal of the American Chemical Society ASAP*
- **Surface-Enhanced Raman Detection of Interfacial Water Enabled by Electrolyte-Specific Reconstruction of Hydrogen-Bond Network Defects in the Electric Double Layer** — score:

1.000 This study examines how the specific interactions of electrolytes at the Ag/electrolyte interface influence the surface-enhanced Raman scattering (SERS) detection of interfacial water. Through in situ electrochemical SERS and molecular dynamics simulations, it reveals that the reconfiguration of hydrogen-bond networks—driven by different cations—affects the O−H spectral response, enhancing the ability to probe changes in hydration networks within electric double layers. Journal: *Wiley: Small Methods: Table of Contents*

- **Decoupling Dissolution from Water Oxidation at Single-Faceted High-Entropy Perovskite Electrodes** — score: 1.000 The study investigates high-entropy perovskite (HEO) electrodes as anodes for the oxygen evolution reaction (OER), focusing on decoupling the processes of electrochemical dissolution and OER. By employing atomic force microscopy and scanning electrochemical cell microscopy, the researchers quantitatively analyze dissolution kinetics and reveal how substitutions in the perovskite structure affect both dissolution and OER charge contributions, leading to insights on the electrodes' intrinsic activity and stability. Journal: *Journal of the American Chemical Society ASAP*
- **Kinetic Control of Polymorphism in Li 10.5 Si 1.5 P 1.5 S 12 –Based Solid Electrolytes** — score: 1.000 This study explores the crystallization behavior of Li10.5Si1.5P1.5S12-based solid electrolytes, highlighting how lithium iodide (LiI) as an additive influences phase transitions between a metastable orthorhombic and a stable tetragonal phase. Through advanced characterization methods, the research reveals insights into the microstructural changes and kinetics of these transitions, underscoring their significance for enhancing the performance of lithium-sulfur solid-state batteries. Journal: *Journal of the American Chemical Society ASAP*
- **Boosting Ionomer Oxygen Transport via Pore Confinement in Carbon Supports for Low-Pt Fuel Cells** — score: 1.000 This study introduces a pore-confinement strategy to enhance the oxygen transport in low-platinum proton exchange membrane fuel cells (PEMFCs) by manipulating the spatial arrangement of platinum nanoparticles within porous carbon supports. By positioning the nanoparticles within mesopores rather than on the surface, the research demonstrates a significant increase in power density due to improved interfacial oxygen transport, offering a novel approach for optimizing low-platinum catalytic systems in fuel cells. Journal: *Journal of the American Chemical Society ASAP*
- **Fluorine-Mediated Molten-Salt Crystallization Regulates Phase Evolution and Structural Stability in Sodium Layered Oxides** — score: 1.000 The article discusses the use of fluorine-assisted molten-salt synthesis to enhance the stability and phase selectivity of Ni- and Co-free P2-type Fe-Mn layered oxides for sodium-ion batteries. This method promotes the formation of large, uniform particles with improved structural coherence and reversible behavior, ultimately reducing crack formation and improving cycling stability compared to conventional synthesis methods. Journal: *Journal of the American Chemical Society ASAP*
- **Regulating Polysulfide Conversion on Fe-N x Single-Atom Catalysts Supported on Graphitic Carbon Nitride Revealed by Semi-In Situ XPS of Li–S Batteries** — score: 1.000 The study investigates how the thermal reduction of urea-derived graphitic carbon nitride supports affects the performance of Fe-Nx single-atom catalysts in lithium-sulfur batteries. By engineering the support before metal incorporation, researchers achieved enhanced polysulfide conversion and improved durability, demonstrating that the local coordination environment significantly influences catalytic efficiency. Additionally, semi-in situ X-ray photoelectron spectroscopy provided insights into sulfur speciation during cycling, revealing accelerated conversion processes facilitated by the modified catalysts. Journal: *ACS Applied Energy Materials ASAP*
- **Elucidating the Role of sALD-Deposited ZrO x Coatings on the Electrochemical Performance of LiNi 0.5 Mn 1.5 O 4 Cathodes** — score: 1.000 The study explores the effectiveness of spatial atomic layer deposition (sALD) in applying nano-ZrOx coatings on LiNi0.5Mn1.5O4 (LNMO) cathodes to enhance their electrochemical performance and stability. Results indicate that an optimal coating thickness of 3.0 nm improves cycling stability by 71% while maintaining rate capability, with reduced interfacial degradation and minimized crosstalk between cathode and anode, suggesting that

both coating thickness and deposition speed significantly influence the longevity of LNMO/graphite cells. Journal: *ACS Applied Energy Materials ASAP*

- **Oxyhalide-Derived Nb-O-Cl Interfacial Frameworks for Mitigating Oxygen-Related Degradation and Facilitating Li+ Transport in High-Voltage Ni-Rich Cathodes** — score: 1.000 The study introduces a reconstructive Nb–O–Cl interfacial framework that enhances the stability and performance of high-voltage Ni-rich cathodes coupled with sulfide electrolytes. This framework mitigates oxygen-related issues like degradation and O2 evolution while facilitating Li+ transport, resulting in reduced interfacial deterioration and significantly improved cycling stability, with an impressive 88.17% capacity retention after 500 cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Dead-Potassium Activation Reconstructing Stabilized Interface Enables Stable Anode-Free Potassium Metal Batteries** — score: 1.000 The article discusses a novel engineering strategy that incorporates bismuth iodide into carbonate-based electrolytes to address the challenges of anode-free potassium metal batteries (AFPMBs). This approach mitigates issues of dead potassium accumulation and dendrite growth by enhancing ion dynamics, leading to improved cycling stability and capacity retention, with cells demonstrating stable performance for over 1300 hours. Journal: *Journal of the American Chemical Society ASAP*
- **Operando Electro-Rheology: Unraveling Decoupled Ion Transport-Mechanical-Morphological Properties in Nanostructured Block Copolymer Electrolytes** — score: 1.000 The article presents a novel operando methodology that integrates rheological and electrochemical measurements to investigate the behavior of LiTFSI-doped polystyrene-block-poly(ethylene oxide) block copolymer electrolytes under dc polarization. The approach reveals the dynamic interplay between ion transport, mechanical response, and morphological changes as the material's structure adapts to varying current densities, demonstrating a significant coupling between interfacial electrochemical processes and mechanical properties. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Mechanistic Insights of Jahn–Teller Effect Suppression in High-Entropy Prussian Blue Analogue Cathodes** — score: 1.000 This study investigates the suppression of the Jahn-Teller effect in high-entropy Prussian blue analogue cathodes, revealing that the typical bond-length splitting associated with J-T distortion is absent, replaced by a broader Mn–N bond-length distribution. Utilizing machine learning and EXAFS, the authors demonstrate that random distortions in the [MnN6] octahedra average out at macroscopic scales, enhancing d-electron delocalization and leading to improved cycling stability for aqueous ammonium-ion batteries, achieving 91% capacity retention over 50,000 cycles. Journal: *Journal of the American Chemical Society ASAP*
- **Quantifying Sodium Metal SEI Dissolution and Relation to Corrosion Using Quartz Crystal Microbalance** — score: 1.000 This study investigates the dissolution of the solid-electrolyte interphase (SEI) in sodium-metal batteries, utilizing quartz crystal microbalance to measure the mass changes of NaPF6 in diglyme under various potential holds. The findings reveal significant SEI dissolution, indicating instability in both organic and inorganic components, which contributes to ongoing corrosion of sodium metal during aging, underscoring the need for improved SEI stability in these battery systems. Journal: *ACS Energy Letters ASAP*
- **High-Entropy-Inspired Trace Multi-cation Doping Enables Structural Reinforcement and Accelerated Na-Ion Transport in Mn-Based Layered Cathodes** — score: 1.000 The study introduces a high-entropy trace doping technique that incorporates five metal elements into P2-Na . MnO to create a new cathode material (HE-NMO) for sodium-ion batteries. This approach improves structural stability and enhances ion transport, leading to significant electrochemical performance with a high discharge capacity of 169.6 mAh g ^{1} and excellent cycle stability. Journal: *ACS Applied Energy Materials ASAP*
- **Resolving Real-Time Interfacial Dynamics on Garnet-Based Solid-State Electrolytes via Snapshot Operando XPS** — score: 1.000 The article presents a novel snapshot operando X-ray photoelectron spectroscopy (XPS) technique that allows for real-time observation of interfacial dynam-

ics in solid-state batteries, particularly during electrochemical lithiation in garnet-based electrolytes. This method, with its enhanced temporal resolution and depth-profiling capabilities, aims to improve the understanding and development of stable electrode interfaces, which are essential for advancing solid-state battery technology. Journal: *Wiley: Small Methods: Table of Contents*

- **Mineral Particle Suspensions in Carbonate Electrolyte Enable LiF-Rich Interphases for Durable Lithium Metal Batteries** — score: 1.000 The study highlights a novel approach to enhance the stability of lithium metal batteries by suspending lithium fluoride (LiF) and lithium oxide (Li2O) particles in a carbonate electrolyte. This method facilitates the preferential adsorption and electronic polarization of fluorinated components, leading to the formation of LiF-rich interphases that effectively suppress dendrite growth and parasitic reactions, achieving significant improvements in cycling performance with 2400 hours of symmetric cell cycling and 85.6% capacity retention after 400 full cell cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Vacancy-Induced Ligand-Hole Character Redirects Oxygen Redox in Li-Rich Layered Oxides** — score: 1.000 The study investigates how preconditioning lithium-rich layered oxides with oxygen vacancies alters their electronic structure, specifically enhancing the ligand-hole character of oxygen. This modification leads to improved charge transfer and limits detrimental structural changes during cycling, resulting in a cathode with reduced voltage decay and enhanced electrochemical performance by promoting more stable oxygen redox behavior. Journal: *Journal of the American Chemical Society ASAP*
- **NaPF 6 -Based Battery Electrolytes near a Sodium Metal Electrode: Role of Organic Solvents on Solid Electrolyte Interphase Formation** — score: 1.000 The study investigates the impact of different organic solvents on the formation of solid electrolyte interphase (SEI) in sodium hexafluorophosphate (NaPF)-based electrolytes used with sodium metal electrodes. Using density functional theory and molecular dynamics simulations, it finds that solvent stability and the reducing properties of the PF anion play crucial roles, with THF and EC:DEC yielding favorable inorganic-rich SEIs characterized by noncovalent Na-F and Na-P bonding, while the behavior of propylene carbonate is distinctly different. Journal: *ACS Applied Energy Materials ASAP*
- **Making 1000 Wh/kg LiNiO 2 Cathode Nearly Strain-Free by Complex-Concentrated Doping** — score: 1.000 The study presents a method for enhancing the performance of LiNiO2 (LNO) cathodes in batteries, achieving nearly strain-free operation through complex-concentrated doping. This approach reduces c-axis strain significantly while maintaining high energy density (973 Wh/kg) and capacity retention after extensive cycling, thereby addressing the challenges posed by chemomechanical degradation in high-energy lithium-ion batteries. Journal: *ACS Energy Letters ASAP*
- **High-Energy and Long-Life Na3V2(PO4)2F3 Cathodes via Multielement V-Site Modulation of Local M–O/F Bonding** — score: 0.900 The article discusses an innovative approach to enhance the Na3V2(PO4)2F3 cathode used in sodium-ion batteries by employing multielement V-site modulation. This strategy significantly improves bonding stability, prevents impurities and fluorine loss, thus achieving a high energy density of 477 Wh kg−1, remarkable cycling stability over 11,000 cycles, and functionality across a wide temperature range from −40 to 80°C, making it suitable for practical applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Water serves as a charge-transfer medium for ultrafast aqueous batteries** — score: 0.900 The article explores the role of interfacial water as a charge-transfer medium in ultrafast aqueous batteries, challenging traditional ion-transport models. It delves into the molecular mechanisms and boundary conditions involved, and discusses the implications for enhancing high-power energy storage in aqueous systems. Journal: *Nature Reviews Materials*
- **Manipulating Li 2 MnO 3 -Like Domain Dispersion in Lithium-Rich Layered Oxides Inherited from Precursor** — score: 0.900 This study explores how the dispersion of Li2MnO3-like domains in lithium-rich layered oxides affects the stability and performance of lithium-ion batteries. The authors demonstrate that the structural characteristics of precursor materials influence the degree of transition-metal clustering, which in turn determines domain formation in the final cathodes. Their

findings reveal that optimized precursor design can enhance electrochemical stability and reduce voltage decay in these battery materials. Journal: *ACS Energy Letters ASAP*

- **Resolving Oxidative and Corrosive Calendar-Aging via Electrolyte Engineering for Stable Lithium Metal Batteries** — score: 0.900 The article explores the use of 1,2-dimethoxypropane (DMP) in localized high-concentration electrolytes (LHCEs) to enhance the lifespan of lithium metal batteries (LMBs) by reducing both lithium metal corrosion and cathode degradation. This innovative approach addresses the dual issues of electrolyte-induced degradation, ultimately achieving an impressive 81% capacity retention over 220 cycles, and emphasizes the importance of mitigating cathode-side degradation for improving battery longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Protonated Porphyrin-Induced Ionic Aggregation Regulation for Optimizing Microphase-Separated Morphology and Performance of Sulfonated Polyimide Proton Exchange Membranes** — score: 0.900 The study introduces a strategy to enhance the performance of sulfonated polyimide proton exchange membranes (PEMs) for direct methanol fuel cells by incorporating protonated porphyrin units into the polymer backbone. This method optimizes ionic aggregation and morphology, resulting in improved proton conductivity (156.7 mS cm–1), dimensional stability, and methanol resistance, particularly in the SPI-TAPP-2.0 membrane, which also demonstrated excellent power density and durability. The success is attributed to the – stacking and acid-base interactions that create efficient proton transport networks. Journal: *ACS Applied Energy Materials ASAP*
- **Stabilizing Interfaces in Magnesium Solid-State Batteries via High-Throughput Coating Design** — score: 0.900 The study investigates the electrochemical stability of magnesium solid electrolytes (SEs) and their compatibility with cathodes for all-solid-state magnesium-ion batteries. By conducting high-throughput screenings of over 19,000 magnesium compounds, the researchers identify seven effective coating materials that can stabilize the battery interfaces, thereby offering valuable strategies to improve battery performance and tackle existing limitations in magnesium battery technology. Journal: *Wiley: Small Methods: Table of Contents*
- **Engineering Advanced Electrocatalysts for Energy-Saving Zinc—Air Hybrid Batteries: Concurrent Energy Storage Coupled With Chemical Upgrading** — score: 0.800 The article provides a comprehensive review of zinc—air hybrid batteries (ZAHBs), emphasizing their potential for efficient energy storage while facilitating chemical upgrading and wastewater treatment. It details advancements in catalyst design through various modulation strategies, summarizes progress in different ZAHB categories, and discusses key challenges and future directions for improving these energy systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Machine Learning in Liquid Electrolyte Design for Rechargeable Batteries: Addressing Fundamental Challenges** — score: 0.800 The article reviews the application of machine learning (ML) in addressing the complex challenges of designing liquid electrolytes for rechargeable batteries, including optimizing ionic transport and electrochemical stability. It outlines specific ML methodologies, dataset construction, and feature representations tailored to electrolyte chemistry, while also identifying future challenges and opportunities to enhance the accuracy and applicability of ML in the field of electrolyte research. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Rational Integration of Multiple Additives for High-Temperature Lithium-Ion Batteries** — score: 0.800 This study explores the interactions between various electrolyte additives in lithium-ion batteries, specifically highlighting the adverse effects of combining succinonitrile (SN) and lithium bis(oxalato)borate (LiBOB) on the electrode's solid electrolyte interphase (SEI) stability. By substituting SN with triisopropanolamine borate (TPAB), the authors demonstrate improved interphase formation and enhanced capacity retention, achieving 68.7% capacity after 300 cycles at high temperatures. Journal: *ACS Energy Letters ASAP*
- **Investigation of Moisture-Induced Degradation Mechanisms and a Reviving Strategy for Sulfide-Based Solid Electrolytes in All-Solid-State Composite Cathode (Adv. Energy Mater. 39/2026)** — score: 0.800 The article examines the mechanisms of moisture-induced degra-

dation in sulfide-based solid electrolytes used in all-solid-state batteries, highlighting how these electrolytes retain reactive surface species that lead to unstable interfaces and degradation with layered oxide cathodes. It also discusses a revival strategy involving O2 annealing that effectively removes these harmful residues, restores the electrolyte surface, and reduces interfacial damage, offering a viable solution for enhancing the performance of moisture-exposed solid electrolytes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Reaction-Driven Coupled Transport–Nanostructure Evolution in SnSb Anodes for High-Energy All-Solid-State Li-Ion Batteries** — score: 0.800 The article discusses the innovative use of a SnSb anode in all-solid-state Li-ion batteries, which addresses common issues such as sluggish transport kinetics and unstable interfaces by facilitating reaction-driven coupled transport and nanostructure evolution. This mechanism enhances electrode performance by maintaining excellent contact with the electrolyte, achieving a high areal capacity of 28.97 mAh cm−2 and ensuring stable cycling beyond 1000 cycles, thus demonstrating significant potential for high-energy density applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Dual Anion-Involved Interfacial Chemistry Enables Ultra-Stable PEO-Based Solid-State Lithium Metal Batteries by Suppressing NMP-Induced Side Reactions** — score: 0.800 The study presents a dual-anion strategy that transforms the role of N-methylpyrrolidone (NMP) from a detrimental side reaction to a beneficial component in solid-state lithium metal batteries. By incorporating the salts LiDFOB and LiNO3, a stable solid electrolyte interphase (SEI) is formed, which enhances lithium ion transport and mitigates side reactions, resulting in improved battery performance with high capacity retention and extended cycling stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Tailoring Crystal Facets Enables Scalable Synthesis of Highly Dispersed Single Crystal NCM90** — score: 0.800 The article presents a novel method for synthesizing highly dispersed single-crystal NCM90 cathodes through surface high-valence metal doping. This technique improves particle dispersion and cycling stability by enhancing metal-oxygen bonding, regulating crystal growth, and exposing favorable lithium migration planes, resulting in a product with higher charging capacity and durability compared to traditional polycrystalline NCM90. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Physically Grounded Descriptors for Machine-Learning-Guided Diluent Design in Lithium Metal Batteries** — score: 0.800 This study presents a novel approach for designing nonfluorinated diluents in lithium metal batteries by using two key descriptors related to electron donation and molecular polarity. By integrating these descriptors with machine learning techniques, the researchers screened over 1.3 million molecules and successfully identified promising candidates that demonstrated stable cycling performance in battery applications, highlighting the importance of controlling specific molecular interactions for effective diluent design. Journal: *Journal of the American Chemical Society ASAP*
- **Localized Cationic Clustering as an Atomic-scale Descriptor for Understanding and Mitigating High-voltage Instability in Ni-rich Layered Oxide Cathodes** — score: 0.800 The article discusses a novel atomic-scale descriptor, localized cationic clustering, which provides insights into high-voltage instability in Ni-rich layered oxide cathodes. By understanding this phenomenon, the research aims to enhance the stability and performance of these materials in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Exploiting the Li–N Interaction of Arginine Additive to In Situ Construct a Cathode Protective Layer for Lean Electrolyte Li–S Batteries** — score: 0.800 This article presents a novel approach to enhance lithium-sulfur (Li-S) battery performance using arginine as an electrolyte additive. The research reveals that arginine interacts with lithium ions and polysulfides to form complexes that contribute to a protective cathode electrolyte interface, minimizing the dissolution of lithium polysulfides and improving sulfur utilization, particularly in lean electrolyte conditions. The results demonstrate a significant improvement in discharge capacity and cycling stability, highlighting the potential for developing high-performance Li-S batteries with lower electrolyte requirements. Journal:

ACS Applied Energy Materials ASAP

- **Interhalogen of Iodine Chloride Accelerates Charge Transfer for Efficient Li 2 O 2 Oxidation in Lithium–Oxygen Batteries** — score: 0.800 The study introduces iodine monochloride (ICl) as an electrochemically generated redox mediator that significantly enhances the oxidation kinetics of Li2O2 in lithium-oxygen batteries. By creating a more polar bond between iodine and chlorine, ICl lowers the energy barrier for Li2O2 decomposition, reducing charge polarization to 0.52 V and extending battery cycling life beyond 800 hours. This advancement offers insights for developing more effective redox mediators in Li-O2 battery technology. Journal: *Journal of the American Chemical Society ASAP*
- **Support Protection Mechanism Enables Efficient and Stable Anion Exchange Membrane Water Electrolyzers** — score: 0.800 The study introduces a support protection mechanism involving lattice-confined ruthenium (Ru) single atoms that enhances the stability and efficiency of Co O -based catalysts in alkaline anion exchange membrane water electrolyzers (AEMWEs). This mechanism reduces cobalt dissolution while optimizing electronic structures for better oxygen evolution reaction (OER) kinetics, achieving a lower overpotential and maintaining stability over 1000 hours. Overall, it presents lattice confinement as a promising strategy for improving catalyst durability and performance in electrochemical applications. Journal: *Journal of the American Chemical Society ASAP*
- **Investigation of IrRuO x -Based Porous Transport Electrodes as a Function of Post-Treatments** — score: 0.800 This study investigates the effects of various post-treatment annealing conditions on electrodeposited IrRuOx porous transport electrodes, focusing on their composition and electrochemical properties. Utilizing X-ray photoelectron spectroscopy and scanning electron microscopy, the findings indicate that different atmospheres during annealing significantly influence the material's stability and catalytic performance, with a comprehensive analysis linking these changes to electrochemical behavior through principal component analysis. Journal: *ACS Applied Energy Materials ASAP*
- **Axial Cl-Coordinated Porous FeCoZn Trimetallic Single-Atom Catalysts: Breaking the Linear Scaling Relation for Superior Oxygen Reduction Reaction and Ultrastable Zinc-Air Batteries** — score: 0.800 This article presents the development of novel porous trimetallic single-atom catalysts (TACs) that enhance oxygen reduction reaction (ORR) performance and improve the stability of zinc-air batteries. Utilizing a unique Fe/Co/Zn composition with axial Zn-Cl coordination, the TACs significantly optimize the adsorption of O2, achieving superior catalytic activity and minimal performance degradation over extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A Tailored Multifunctional Lithium Organosilicate Salt Constructing Robust Interphases for High Voltage Lithium Metal Batteries** — score: 0.800 The article presents the synthesis of a novel lithium organosilicate salt, lithium fluorobis(fluoropinacolatoborato)silicate (LiPBS), which forms effective interphases on both the cathode and anode of lithium metal batteries. LiPBS enhances cycling stability and performance by creating protective layers that prevent oxidative decomposition and effectively scavenge harmful substances, leading to improved capacity retention under high-voltage and elevated temperature conditions compared to traditional electrolytes like LiPF6. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Tracing and Healing Multiscale Defects for Cost-Effective Lithium Carbonate-Derived Layered Oxide Cathodes** — score: 0.800 This study highlights the performance shortcomings of lithium carbonate-derived layered oxide cathodes compared to those made from lithium hydroxide. It identifies multiscale defects caused by the carbonate synthesis process and introduces a single secondary heat treatment that effectively repairs these defects, achieving performance comparable to hydroxide-based materials while reducing production costs by 9% to 16%. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Low-Cost Iron/Manganese-Based Polyanionic Cathodes for Sustainable Na-Ion Batteries** — score: 0.800 The article reviews iron and manganese-based polyanionic cathodes for sustainable

sodium-ion batteries (NIBs), highlighting their cost-effectiveness, high voltage, and cycling stability. It discusses material design, advancements, and challenges while analyzing the structure-function relationships through various synthesis methods and electrochemical behaviors, aiming to guide future NIB development. Journal: *ACS Energy Letters ASAP*

- **Insoluble –d -Conjugated Small Molecules Featuring p -Type Ligands Enable All-Organic Dual-Ion Batteries** — score: 0.800 This article presents a novel approach to developing all-organic dual-ion batteries using insoluble –d conjugated small molecules featuring p-type triphenylamine-derived ligands. By utilizing a coordination-driven strategy, these molecules are thermodynamically stabilized against dissolution while achieving high electronic conductivity and capacity. The resulting battery demonstrates excellent performance with a voltage of 2.4 V and over 1000 cycles, highlighting the potential of this method in advancing organic battery technology. Journal: *ACS Energy Letters ASAP*
- **Operando Time-Resolved Spectroelectrochemical Microscopy Reveals Distinct Electron Depletion Pathways in Electrocatalysis** — score: 0.800 The study presents a novel operando time-resolved spectroelectrochemical microscopy technique that allows for the real-time mapping of electron density in electrocatalysts. Using MoS2, the researchers found that electron depletion occurs at active sites during hydrogen evolution, revealing two distinct mechanisms: reaction-limited at active edges and injection-limited in bilayer areas. This work enhances the understanding of the relationship between electron dynamics and catalytic efficiency, providing new insights for electrocatalysis research. Journal: *Journal of the American Chemical Society ASAP*
- **Weak-Solvating Carboxylate Ester Electrolyte Enables Hybrid Interphase via Cooperative Additives toward Wide-Temperature Sodium Metal Batteries** — score: 0.800 The article presents a novel electrolyte system for sodium metal batteries that combines methyl propanoate with VC and NaDFOB additives, enhancing both Na+ transport and interfacial stability across various temperatures. This strategy results in significant improvements in ionic conductivity, long-term cycling stability, and capacity retention, demonstrating its potential for practical applications in energy storage systems. Journal: *ACS Energy Letters ASAP*